

# Cognifyz Technologies

*Data Analysis Internship*

## Level 3

Task 1: Restaurant Reviews | Task 2: Votes Analysis | Task 3: Price Range vs. Services

*Dataset: Zomato Restaurant Dataset (9,551 restaurants)*

## Task 1: Restaurant Reviews

**Objective:** Analyze text reviews for common positive/negative keywords and the relationship between review length and rating.

### Data Limitation

The provided dataset does not contain a free-text review column — it does not include individual written reviews, so keyword extraction and review-length analysis in the literal sense are not possible with this data. The dataset does, however, include a categorical 'Rating text' field (e.g. Excellent, Very Good, Good, Average, Poor, Not rated), which serves as a useful proxy for review sentiment.

### Rating Text (Sentiment Proxy) Distribution

| Rating Text | Count |
|-------------|-------|
| Average     | 3737  |
| Not rated   | 2148  |
| Good        | 2100  |
| Very Good   | 1079  |
| Excellent   | 301   |
| Poor        | 186   |

Insight: 'Average' is the single most common sentiment label, followed by 'Not rated' and 'Good'. Positive labels ('Very Good' + 'Excellent' = 1,380) outnumber the negative label 'Poor' (186) by a wide margin, indicating that among restaurants that do receive ratings, sentiment skews positive. To perform genuine keyword and review-length analysis, a dataset with raw review text (e.g. Yelp or Zomato review exports) would be required.

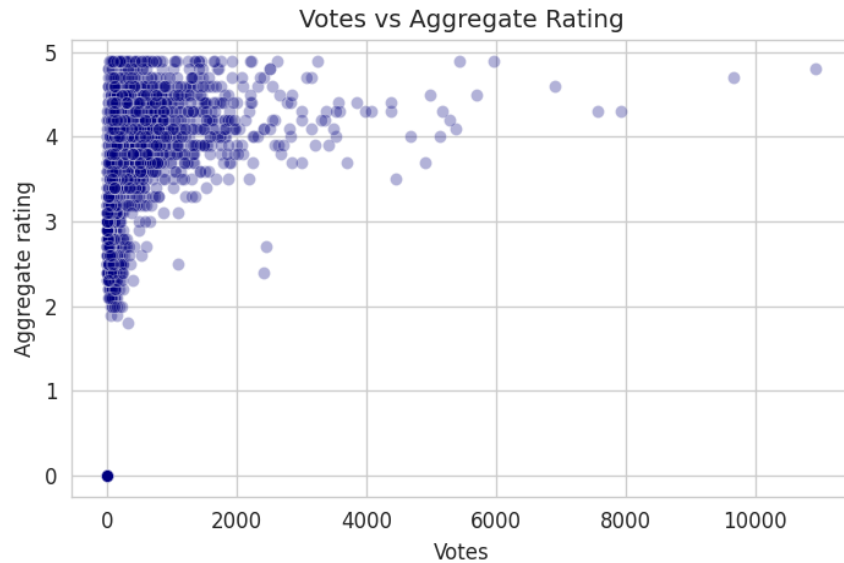
## Task 2: Votes Analysis

**Objective:** Identify restaurants with the highest/lowest votes and analyze the correlation between votes and rating.

### Result

| Metric        | Restaurant                                    | Votes  | Rating |
|---------------|-----------------------------------------------|--------|--------|
| Highest Votes | Toit                                          | 10,934 | 4.8    |
| Lowest Votes  | Cantinho da Gula (and other tied restaurants) | 0      | N/A    |

Correlation coefficient (Votes vs. Aggregate Rating): 0.314 (weak-to-moderate positive correlation)



Insight: There is a weak-to-moderate positive correlation between number of votes and rating — highly-voted restaurants tend to skew toward higher ratings, but the relationship is far from strict. Many restaurants with low vote counts still receive high ratings, and popularity (vote count) alone is not a reliable predictor of quality.

### Task 3: Price Range vs. Online Delivery and Table Booking

**Objective:** Analyze the relationship between price range and availability of online delivery / table booking, and test whether higher-priced restaurants are more likely to offer these services.

#### % Offering Online Delivery, by Price Range

| Price Range | No Delivery % | Delivery % |
|-------------|---------------|------------|
| 1 (Low)     | 84.2%         | 15.8%      |
| 2           | 58.7%         | 41.3%      |
| 3           | 70.8%         | 29.2%      |
| 4 (High)    | 91.0%         | 9.0%       |

#### % Offering Table Booking, by Price Range

| Price Range | No Booking % | Booking % |
|-------------|--------------|-----------|
| 1 (Low)     | 100.0%       | 0.0%      |
| 2           | 92.3%        | 7.7%      |
| 3           | 54.3%        | 45.7%     |
| 4 (High)    | 53.2%        | 46.8%     |



Insight: The two services show opposite patterns. Table booking increases steadily and substantially with price range — nearly half of price-range-4 restaurants offer table booking versus none in price-range-1 — consistent with the expectation that higher-end restaurants are more likely to offer reservations. Online delivery, however, peaks at price range 2 (mid-low) and is lowest at price range 4, suggesting premium/fine-dining restaurants prioritize in-house dining experience and table service over delivery. So higher-priced restaurants are more likely to offer table booking, but not more likely to offer online delivery.